

A Projection Theorem for Cylindrical Algebraic Decomposition via Elimination Ideals

An informal proof extracted from a Lean 4 formalization

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Abstract

McCallum’s projection theorem (Theorem 3.2.3 of his 1985 thesis) is the mathematical foundation of the McCallum projection operator for cylindrical algebraic decomposition (CAD): if every coefficient, discriminant and pairwise resultant arising from a squarefree basis A is order-invariant on a connected analytic submanifold $S \subseteq \mathbb{R}^{r-1}$, then the elements of A are analytically delineable on S with pairwise disjoint, order-invariant sections. We prove a generalization in which the discriminants and resultants are replaced by *arbitrary nonzero elements of the corresponding elimination ideals* $\langle F, \partial F / \partial x_r \rangle \cap \mathbb{R}[x]$ and $\langle F, G \rangle \cap \mathbb{R}[x]$. The motivation is proof reconstruction: a verifier checking a CAD-based certificate need not recompute resultants and discriminants—membership of the projection polynomials in the elimination ideals can be certified by Bézout cofactors and checked by polynomial arithmetic alone.

The proof follows the architecture of McCallum’s original argument and reuses its intermediate results—complexification, Weierstrass preparation, Zariski’s equisingularity theorem (Theorem 4.1.1 of the thesis), and the return to the real domain—*verbatim*. Exactly three ingredients change: (i) the identity $D = \prod_i \text{disc}(F_i) \cdot \prod_{i < j} \text{res}(F_i, F_j)^2$ (thesis Theorem 2.3.3), used to deduce the projection theorem from the lifting theorem, is replaced by a purely ring-theoretic *radical lemma*: some power of the product of the chosen elimination-ideal elements again lies in the elimination ideal of the product; (ii) the factorization $E = Q \cdot \text{disc}(h)$ of the discriminant across Weierstrass preparation (thesis equation (3.3.8), again via Theorem 2.3.3) is replaced by a *norm-resultant identity*: any P in $\langle h, h' \rangle \cap \mathcal{O}$ for a Weierstrass polynomial h of degree m over the ring $\mathcal{O}$ of analytic germs satisfies $P^m = \pm \text{disc}(h) \cdot Q$ with $Q \in \mathcal{O}$; (iii) the Bézout cofactors witnessing $P \in \langle f, f' \rangle$ must be transported through the coordinate change, the complexification and the Weierstrass preparation, a bookkeeping obligation absent from the original proof. Everything else, including the entire Chapter 4 of the thesis (the Zariski theorem), is used unchanged.

The complete proof, including a self-contained proof of Zariski’s theorem and of the Weierstrass preparation and division theorems, has been formalized and machine-checked in Lean 4 over Mathlib; the main theorem (`mccallum_3_2_3_generalized`) depends on no axioms beyond Lean’s standard three (`propext`, `Classical.choice`, `Quot.sound`). This document is the informal rendering of that formal proof.

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1 Introduction

McCallum’s improved projection operator [1, 2] replaces the Collins projection in CAD construction by the much smaller set

$$P(A) = \text{coeff}(A) \cup \text{disc}(A) \cup \text{res}(A),$$

and his Theorem 3.2.3 shows that *order*-invariance of $P(A)$ on a cell lifts to delineability and order-invariance of A over the cell. The discriminants and resultants enter the proof through exactly one interface: $\text{disc}(F)$ (resp. $\text{res}(F, G)$) is a nonzero polynomial in the base variables $x = (x_1, \dots, x_{r-1})$ that vanishes wherever $F, \partial F/\partial x_r$ (resp. F, G) have a common root—that is, a nonzero element of an *elimination ideal*. This suggests the generalization proved here: the theorem survives, with the same conclusions, when each $\text{disc}(F)$ is replaced by an arbitrary chosen nonzero $d(F) \in \langle F, \partial F/\partial x_r \rangle \cap \mathbb{R}[x]$ and each $\text{res}(F, G)$ by an arbitrary chosen nonzero $r(F, G) \in \langle F, G \rangle \cap \mathbb{R}[x]$.

The practical motivation comes from certificate checking. An untrusted CAD implementation can emit, along with its answer, the projection polynomials it used together with Bézout cofactors witnessing the memberships $d(F) = AF + B \partial F/\partial x_r$ and $r(F, G) = AF + BG$. A verifier then only needs to check these polynomial identities—no resultant, subresultant or discriminant computation, and no commitment to any particular projection implementation. The generalized theorem is exactly the statement that such certificates are sound.

Sections 2–3 fix notation and state the two main results, the generalized lifting theorem (Theorem 3.3, “3.2.1’”) and the generalized projection theorem (Theorem 3.5, “3.2.3’”). Section 4 develops the three new ingredients. Sections 5 and 6 give the proofs, following McCallum’s §3.2–3.3 step by step and marking each step **[unchanged]** or **[changed]** relative to the thesis. Section 7 recalls the Zariski theorem (thesis Chapter 4), which is used as a black box with *no* modification, and records how the formalization proves it. Section 8 summarizes the correspondence between this document and the Lean development.

Throughout, thesis references “Theorem *m.n.k*” are to McCallum’s 1985 PhD thesis [1].

2 Preliminaries and notation

All background is as in Chapter 2 of the thesis and is used without change. We single out what is needed, with thesis numbering: analytic functions and convergent power series (§2.1); the complexification of a real-analytic function by its power series (Theorem 2.1.2); the implicit mapping theorem (Theorem 2.1.7); the Weierstrass preparation theorem (Theorem 2.1.10); analytic submanifolds and straightening coordinate systems (Theorem 2.2.1); the behaviour of the vanishing order under analytic isomorphisms (Theorem 2.3.1); the root continuity principle (Theorem 2.3.2); and the discriminant-of-a-product formula (Theorem 2.3.3), which we will use *only* inside the proof of the Zariski theorem, not in the main argument.

Throughout, $x = (x_1, \dots, x_{r-1})$, and a polynomial $f(x, x_r) \in \mathbb{R}[x, x_r]$ is regarded as univariate in x_r over $\mathbb{R}[x]$. We write $f' = \partial f / \partial x_r$. For $a \in \mathbb{R}^{r-1}$, $f(a, x_r) \in \mathbb{R}[x_r]$ denotes the specialization.

Definition 2.1 (Order; order-invariance (thesis §2.3, §3.2)). For $K = \mathbb{R}$ or $\mathbb{C}$, an analytic function F defined near $p \in K^n$, the *order* $\text{ord}_p F \in \mathbb{N} \cup \{\infty\}$ is the least k such that some partial derivative of F of total order k does not vanish at p (∞ if all vanish). F is *order-invariant* in a subset S of its domain if $p \mapsto \text{ord}_p F$ is constant on S . For a polynomial $g \in \mathbb{R}[x]$ this applies with $F = g$ as a function on $\mathbb{R}^{r-1}$; for $f \in \mathbb{R}[x, x_r]$, order-invariance “in a subset of $\mathbb{R}^r$ ” refers to f as a function of all r variables.

Note that the order at a point of S is always the *ambient* order—the order of F as a function on a neighbourhood in K^n , not of the restriction $F|_S$ (which is identically zero in the interesting cases).

Definition 2.2 (Analytic submanifold; analytic delineability (thesis §2.2, §3.2)). $S \subseteq \mathbb{R}^{r-1}$ is a (nonempty, s -dimensional) *analytic submanifold* if every $p \in S$ has an open neighbourhood W and an analytic $F : W \rightarrow \mathbb{R}^{r-1-s}$, submersive at p , with $S \cap W = F^{-1}(0) \cap W$. A polynomial $f(x, x_r)$ is *analytically delineable* on S if there are $k \geq 0$ analytic functions $\theta_1 < \dots < \theta_k$ on S such that, for every $a \in S$, the real roots of $f(a, x_r)$ are exactly $\theta_1(a), \dots, \theta_k(a)$, and the multiplicity of $\theta_i(a)$ as a root of $f(a, x_r)$ is a constant $m_i \geq 1$ independent of a . The graphs of the θ_i are the *f-sections* over S .

Definition 2.3 (Squarefree basis). A finite set $A \subseteq \mathbb{R}[x, x_r]$ is a *squarefree basis* if every element has positive degree in x_r , is squarefree, and the elements are pairwise coprime.

3 The generalized theorems

Definition 3.1 (Generalized reduced projection). Let A be a squarefree basis. A *generalized reduced projection* of A is any set

$$P'(A) = \text{coeff}(A) \cup \{d(F) : F \in A\} \cup \{r(F, G) : F, G \in A, F \neq G\},$$

where $\text{coeff}(A)$ is the set of nonzero x_r -coefficients of elements of A , and $d(F) \in \langle F, F' \rangle \cap \mathbb{R}[x]$, $r(F, G) \in \langle F, G \rangle \cap \mathbb{R}[x]$ are *chosen nonzero* elements of the indicated elimination ideals.

Remark 3.2 (Non-vacuity; recovering McCallum’s $P(A)$). The resultant lies in the ideal generated by its arguments (Sylvester matrix), so $\text{res}(F, F') \in \langle F, F' \rangle \cap \mathbb{R}[x]$ and $\text{res}(F, G) \in \langle F, G \rangle \cap \mathbb{R}[x]$; moreover $\text{res}(F, F') = \pm \text{lc}(F) \cdot \text{disc}(F)$. Taking $d(F) = \text{res}(F, F')$ and $r(F, G) = \text{res}(F, G)$, these are nonzero exactly under the squarefree-basis hypotheses, and they are order-invariant on S whenever the elements of McCallum’s $P(A)$ are (orders add over products, and $\text{lc}(F) \in \text{coeff}(A)$). Hence Theorem 3.5 below subsumes thesis Theorem 3.2.3. Conversely, a nonzero element of $\langle F, F' \rangle \cap \mathbb{R}[x]$ exists *only* if F is squarefree: a repeated factor of F divides

every element of $\langle F, F' \rangle$ in $\mathbb{R}(x)[x_r]$ and would force the elimination ideal to vanish. The membership hypotheses below therefore silently carry the nondegeneracy (squarefreeness, $\text{disc} \neq 0$) assumptions of the original statements.

Theorem 3.3 (Generalized lifting theorem, “3.2.1’”). *Let $f(x, x_r) \in \mathbb{R}[x, x_r]$ and let S be a connected analytic submanifold of $\mathbb{R}^{r-1}$ on which f is degree-invariant and such that $f(a, x_r) \neq 0$ for every $a \in S$. Suppose there exists $P \in \mathbb{R}[x]$ with*

$$P \neq 0, \quad P \in \langle f, f' \rangle \cap \mathbb{R}[x], \quad P \text{ order-invariant in } S.$$

Then f is analytically delineable on S , and for every continuous root function θ of f on S (i.e. $f(a, \theta(a)) = 0$ for all $a \in S$), f is order-invariant in the graph of θ . In particular f is order-invariant in each f -section over S .

Remark 3.4. Thesis Theorem 3.2.1 is the special case $P = D = \text{disc}(f)$ (more precisely $P = \text{res}(f, f')$, cf. Remark 3.2); its hypotheses “ f squarefree with $D \neq 0$ ” are subsumed by $P \neq 0$ as explained above. The conclusion quantified over arbitrary continuous root functions is the form delivered by the formal proof; it implies (and, given delineability, is equivalent to) order-invariance in each section.

Theorem 3.5 (Generalized projection theorem, “3.2.3’”). *Let A be a nonempty finite squarefree basis of r -variate polynomials, $r \geq 2$, and let S be a connected analytic submanifold of $\mathbb{R}^{r-1}$. Suppose each element of A is not identically zero on S , and each element of some generalized reduced projection $P'(A)$ (Definition 3.1) is order-invariant in S . Then each element of A is degree-invariant and analytically delineable on S , the sections of A over S are pairwise disjoint, and each element of A is order-invariant in every section of A over S .*

4 The three new ingredients

4.1 A norm-resultant identity for monic polynomials

In McCallum’s proof of the lifting theorem, the discriminant E of the transformed polynomial g is related to the discriminant of its Weierstrass polynomial h by the explicit factorization $E = Q \cdot \text{disc}(h)$, obtained from the discriminant-of-a-product formula (thesis Theorem 2.3.3, eq. (3.3.8)). For a general witness $P \in \langle f, f' \rangle \cap \mathbb{R}[x]$ no such product formula exists. One might hope that $\langle h, h' \rangle \cap R = (\text{res}(h, h'))$ for monic h over a ring R , which would give $\text{disc}(h) \mid P$ directly; this is **false**:

$$R = \mathbb{Q}[t], \quad h = z^3 + t, \quad g = h' = 3z^2 : \quad 3t = 3 \cdot h - z \cdot g \in \langle h, g \rangle \cap R, \quad \text{res}(h, g) = 27t^2 \nmid 3t. \quad (1)$$

The correct statement goes through the algebra norm and costs an m -th power.

Lemma 4.1 (Norm-resultant identity). *Let R be a commutative ring and $h \in R[z]$ monic of degree m . For every $g \in R[z]$, the norm of the multiplication-by- $\bar{g}$ endomorphism of the free rank- m R -module $R[z]/(h)$ equals the resultant:*

$$N_{R[z]/(h)/R}(\bar{g}) = \text{res}(h, g).$$

Proof. Both sides are compatible with base change along ring maps (for the norm, because the matrix of multiplication by $\bar{g}$ in the basis $1, \bar{z}, \dots, \bar{z}^{m-1}$ has entries the coefficients of $g z^j \bmod h$, and reduction mod a monic polynomial commutes with ring maps; for the resultant, by the standard $\text{res}(h^\varphi, g^\varphi) = \varphi(\text{res}(h, g))$ for monic h). By a universal-coefficients reduction it therefore suffices to prove the identity when $R = K$ is a field over which h splits, $h = \prod_{i=1}^m (z - \alpha_i)$. There $\text{res}(h, g) = \prod_i g(\alpha_i)$, and the norm factors the same way: by the multiplicativity of determinants along the filtration induced by $K[z]/(h) \twoheadrightarrow K[z]/(h/(z - \alpha_1))$, whose kernel is a line on which $\bar{g}$ acts by the scalar $g(\alpha_1)$, induction on m gives $N(\bar{g}) = \prod_i g(\alpha_i)$. $\square$

Corollary 4.2 (Norm identity for elimination ideals). *Let R be a commutative ring, $h \in R[z]$ monic of degree m , $g \in R[z]$, and let $P \in R$ satisfy $P \in \langle h, g \rangle \cap R$, say $P = ah + bg$ with $a, b \in R[z]$. Then*

$$P^m = \text{res}(h, g) \cdot Q, \quad Q = N_{R[z]/(h)/R}(\bar{b}) \in R.$$

In particular, for $g = h'$: $P^m = \pm \text{disc}(h) \cdot Q$.

Proof. Reducing $P = ah + bg$ modulo h gives $\bar{g}\bar{b} = P$ in $R[z]/(h)$. Apply the norm: it is multiplicative, $N(\bar{g}) = \text{res}(h, g)$ by Lemma 4.1, and $N(P) = P^m$ since the scalar P acts as $P \cdot \text{id}$ on a free module of rank m . For $g = h'$ use $\text{res}(h, h') = \pm \text{disc}(h)$ (h monic). $\square$

Two features of Corollary 4.2 matter in the application. First, it holds over an *arbitrary* commutative ring; we will apply it with $R = \mathcal{O}$ the ring of germs at 0 of holomorphic functions on $\mathbb{C}^{r-1}$, so that the cofactor $Q = N(\bar{b})$ is automatically a *holomorphic germ*—being polynomial in the coefficients of b and h —provided the Bézout cofactor b has analytic coefficients. This is why the membership $P \in \langle h, h' \rangle$ must be transported with analytic cofactors (§6.3). Second, the exponent m is harmless: the order of P^m at a point is $m \cdot \text{ord } P$, so constant order of P along a set gives constant order of P^m . In the counterexample (1): $(3t)^3 = 27t^3 = \text{res}(h, g) \cdot t$, with $Q = t$.

4.2 Order-invariance of factors: Lemma 3.2.2 in ambient form

The second ingredient is the “only if” half of thesis Lemma 3.2.2 (whose proof the thesis records as “straightforward”), in precisely the ambient, along-a-subset form in which the thesis itself applies it, and with the non-degeneracy hypotheses weakened from “each factor nonconstant” to “each factor not identically zero”. We include the complete proof, since the statement is applied in a situation where both factors vanish identically *on* the subset S (only their ambient orders are finite), and since the proof is genuinely where connectedness of S enters.

Lemma 4.3 (Order-invariance descends to factors). *Let $U \subseteq \mathbb{C}^n$ be a connected open set, $f, g : U \rightarrow \mathbb{C}$ holomorphic and neither identically zero on U . Let $S \subseteq U$ be preconnected, let $z_0 \in S$, and suppose $\text{ord}_z(fg) = \text{ord}_{z_0}(fg)$ for all $z \in S$. Then $\text{ord}_z f = \text{ord}_{z_0} f$ and $\text{ord}_z g = \text{ord}_{z_0} g$ for all $z \in S$.*

Proof. Since $f \not\equiv 0$ on the connected open U , the identity theorem gives $\text{ord}_z f < \infty$ for *every* $z \in U$; likewise for g . The order at a point is a valuation: $\text{ord}_z(fg) = \text{ord}_z f + \text{ord}_z g$. Set $a = \text{ord}_{z_0} f$, $b = \text{ord}_{z_0} g$, $c = a + b = \text{ord}_{z_0}(fg)$; all are finite, and $\text{ord}_z f + \text{ord}_z g = c$ for all $z \in S$.

Consider $A = \{z \in S : \text{ord}_z f > a\}$. Because the sum of orders is c and all orders are finite, $A = \{z \in S : \text{ord}_z g < b\}$. The first description shows A is closed in S ($\text{ord}_z f \geq a + 1$ says that all partial derivatives of f of total order $\leq a$ vanish at z , a closed condition); the second shows A is open in S ($\text{ord}_z g \leq b - 1$ says that some particular partial derivative of order $\leq b - 1$ is nonzero at z , an open condition). Thus A is clopen in the preconnected S and misses z_0 , hence $A = \emptyset$. By symmetry $\{z \in S : \text{ord}_z g > b\} = \emptyset$ as well, and since the orders sum to c everywhere on S , $\text{ord } f \equiv a$ and $\text{ord } g \equiv b$ on S . $\square$

Remark 4.4. The hypotheses concern the *ambient* orders at points of S and ambient non-vanishing ($f \not\equiv 0$ on U , not on S). This is essential: in the application $f = \text{disc}(h)$ restricted to the section T^* is identically zero whenever the tracked root has multiplicity $m \geq 2$ (for instance $f = x_3^2 - x_2$ gives a Weierstrass polynomial with $\text{disc}(h) = 4x_2$, identically zero on $T^* = \{x_2 = 0\}$, of constant ambient order 1 there). Finiteness of the ambient orders comes from the identity theorem on U , never from non-vanishing on S .

4.3 The radical lemma

The third ingredient replaces the use of Theorem 2.3.3 ($D = \prod \text{disc}(F_i) \cdot \prod \text{res}(F_i, F_j)^2$) in the deduction of the projection theorem from the lifting theorem. Its content: the product of the chosen elimination-ideal elements, raised to a suitable power, is itself an elimination-ideal element for the product polynomial. The proof is purely ring-theoretic—no algebraically closed field, no Nullstellensatz, no descent of coefficients—using only the characterization of the radical of an ideal as the intersection of the primes containing it.

Lemma 4.5 (Radical lemma). *Let R_0 be a commutative ring, let $F_1, \dots, F_n \in R_0[x_r]$, and set $f = \prod_{i=1}^n F_i$. Suppose given*

$$d_i \in \langle F_i, F'_i \rangle \cap R_0 \quad (1 \leq i \leq n), \quad r_{ij} \in \langle F_i, F_j \rangle \cap R_0 \quad (i \neq j),$$

and set $\mathcal{R} = \prod_i d_i \cdot \prod_{i \neq j} r_{ij} \in R_0$. Then $\mathcal{R}^N \in \langle f, f' \rangle \cap R_0$ for some $N \geq 1$.

Proof. It suffices to show $\mathcal{R} \in \sqrt{\langle f, f' \rangle}$ in $R_0[x_r]$, since then $\mathcal{R}^N \in \langle f, f' \rangle$ for some N , and $\mathcal{R}^N \in R_0$. As $\sqrt{I} = \bigcap \{\mathfrak{p} \text{ prime} : \mathfrak{p} \supseteq I\}$, fix a prime $\mathfrak{p} \supseteq \langle f, f' \rangle$ of $R_0[x_r]$ and show $\mathcal{R} \in \mathfrak{p}$.

From $f = \prod_i F_i \in \mathfrak{p}$ and primality, some $F_{i_0} \in \mathfrak{p}$. Write $f = F_{i_0} \cdot G$ with $G = \prod_{i \neq i_0} F_i$; then $f' = F'_{i_0} G + F_{i_0} G'$, and since $f' \in \mathfrak{p}$ and $F_{i_0} G' \in \mathfrak{p}$, we get $F'_{i_0} G \in \mathfrak{p}$. By primality, either $F'_{i_0} \in \mathfrak{p}$, in which case $\langle F_{i_0}, F'_{i_0} \rangle \subseteq \mathfrak{p}$ and hence $d_{i_0} \in \mathfrak{p}$; or some $F_{j_0} \in \mathfrak{p}$ with $j_0 \neq i_0$, in which case $\langle F_{i_0}, F_{j_0} \rangle \subseteq \mathfrak{p}$ and hence $r_{i_0 j_0} \in \mathfrak{p}$. Either way one factor of $\mathcal{R}$ lies in $\mathfrak{p}$, so $\mathcal{R} \in \mathfrak{p}$. $\square$

Remark 4.6. Geometrically, $\mathcal{R}$ vanishes on the projection of the repeated-root locus of f , which decomposes as $V(f, f') = \bigcup_i V(F_i, F'_i) \cup \bigcup_{i < j} V(F_i, F_j)$; the lemma is the algebraic shadow of that decomposition. The prime-ideal argument above proves it directly over $R_0 = \mathbb{R}[x]$ (indeed over any commutative ring), avoiding both the passage to $\overline{K}$ and the descent of the membership back to $\mathbb{R}[x]$ that a Nullstellensatz argument would require. The exponent N is immaterial for order-invariance, by additivity of orders.

5 Proof of the projection theorem from the lifting theorem

This mirrors the thesis proof of Theorem 3.2.3 (§3.2), with Lemma 4.5 substituting for Theorem 2.3.3.

Proof of Theorem 3.5. Degree-invariance of each $F \in A$ on S follows from order-invariance of its coefficients exactly as in the thesis: the leading coefficient is a nonzero polynomial of constant order on S ; if it vanished at some $a \in S$ its order there would be ≥ 1 , hence ≥ 1 everywhere on S , i.e. it would vanish identically on S ; applying this argument to successive coefficients and using that F is not identically zero on S pins the specialized degree. **[unchanged]**

Disjointness of sections is immediate from pairwise coprimality: a Bézout identity $uF + vG = 1$ in $\mathbb{R}(x)[x_r]$, cleared of denominators and specialized at $a \in S$, prevents common roots of $F(a, x_r)$ and $G(a, x_r)$. **[unchanged]**

Let $f = \prod_{F \in A} F$; it is squarefree (the F are squarefree and pairwise coprime), of positive degree, degree-invariant on S , and $f(a, x_r) \neq 0$ for $a \in S$. **[unchanged]**

[changed] *The witness.* Let $\mathcal{R} = \prod_F d(F) \cdot \prod_{F \neq G} r(F, G) \in \mathbb{R}[x]$. Each factor is nonzero, so $\mathcal{R} \neq 0$; each factor is order-invariant in S , so $\mathcal{R}$ is order-invariant in S (orders add over products). By Lemma 4.5 there is $N \geq 1$ with $P := \mathcal{R}^N \in \langle f, f' \rangle \cap \mathbb{R}[x]$; and $P \neq 0$ is order-invariant in S since $\text{ord}_a P = N \text{ord}_a \mathcal{R}$. (In the thesis, this paragraph is the computation $D = \prod \text{disc}(F_i) \prod \text{res}(F_i, F_j)^2$ followed by Lemma 3.2.2.)

Applying Theorem 3.3 to f with witness P : f is analytically delineable on S and order-invariant in the graph of every continuous root function of f on S . Delineability and the root-multiplicity structure pass from the product f to each factor $F \in A$ as in the thesis (each

root of $F(a, \cdot)$ is a root of $f(a, \cdot)$; on sections, the multiplicities of the coprime factors do not interact); and on the graph of a continuous root function θ of any $G \in A$ (a continuous root function of f as well), the order-invariance of the product f descends to each factor F by Lemma 3.2.2 (in the ambient form, Lemma 4.3, applied to F and its cofactor in f , both of which specialize nontrivially at every point of S). **[unchanged]** $\square$

6 Proof of the lifting theorem

We follow the thesis proof of Theorem 3.2.1 (§3.3) step by step. Recall its architecture: the theorem reduces to a local statement (Assertion 1), which in turn reduces to a nonsplitting statement about each individual real root (Assertion 2); Assertion 2 splits into the case $s = r - 1$ (S open) and the case $1 \leq s \leq r - 2$ (positive codimension), and the latter proceeds by coordinate change, complexification, Weierstrass preparation, the Zariski theorem, and a return to the real domain.

6.1 Reduction to the local statements

[unchanged] The deduction of the theorem from Assertion 1 (local delineability: each $p \in S$ has a neighbourhood N with f analytically delineable on $S \cap N$ and order-invariant in each section over $S \cap N$) is exactly as on thesis pp. 22–23: by connectedness of S the number of distinct real roots, and the list of their multiplicities, are locally constant hence constant; the local analytic root functions agree on overlaps (they enumerate the same roots in increasing order) and patch to global ones; order-invariance globalizes by a clopen argument on S . The reduction of Assertion 1 to Assertion 2 (nonsplitting of each root cluster, eq. (3.3.1)) via the root continuity principle (Theorem 2.3.2)—choosing ϵ below a third of the minimal root separation of $f(p, x_r)$, circles C_i around the distinct complex roots α_i of multiplicity m_i , and a neighbourhood N_0 on which each C_i encloses exactly m_i roots—is also unchanged, as is the translation reducing each real root to $\alpha_i = 0$. (The translation $x_r \mapsto x_r + \alpha_i$ fixes $\mathbb{R}[x]$, so it transports the witness P and its Bézout cofactors verbatim.)

For the rest of the section, fix $p \in S$, a real root $\alpha = 0$ of $f(p, x_r)$ of multiplicity $m \geq 1$, and the data ϵ , C , N_0 as above. We must produce a neighbourhood $N \ni p$ and an analytic $\theta : S \cap N \rightarrow (-\epsilon, \epsilon)$ with: $f(x, \alpha) = 0 \iff \alpha = \theta(x)$ for $x \in S \cap N$, $|\alpha| < \epsilon$; and f order-invariant in the graph of θ .

6.2 Case $s = r - 1$ (S open)

[changed] (simplified). Since S is open and connected and P is a nonzero polynomial of constant order on S , P vanishes nowhere on S : a nonzero polynomial cannot vanish identically on a nonempty open set, so its order is 0 at some point of S , hence at every point. Now write the membership as a Bézout identity

$$P(x) = A(x, x_r) f(x, x_r) + B(x, x_r) f'(x, x_r)$$

and specialize at any $a \in S$: since $P(a) \neq 0$, the univariate polynomials $f(a, x_r)$ and $f'(a, x_r)$ generate the unit ideal, i.e. $f(a, x_r)$ is *separable*. In particular $m = 1$ and $f'(p, 0) \neq 0$, and the implicit function theorem (Theorem 2.1.7) yields the analytic simple-root function θ near p with f of constant order 1 on its graph ($f' \neq 0$ there). This proves Assertion 2 with constant section order 1.

(The thesis argument passes through the identity $(-1)^{m(m-1)/2} R(x) = f_0(x) D(x)$ and the non-vanishing of the leading coefficient to conclude $R(p) \neq 0$; the elimination-ideal hypothesis makes this detour unnecessary.)

6.3 Case $1 \leq s \leq r - 2$: coordinate change and transport of the witness

Coordinate change. **[unchanged]** By Theorem 2.2.1 choose a connected neighbourhood $U \subseteq N_0$ of p and an analytic coordinate system $\Phi : U \rightarrow V$ with $S \cap U = \{x \in U : \phi_{s+1}(x) = \cdots = \phi_{r-1}(x) = 0\}$; let $T = \{y : y_{s+1} = \cdots = y_{r-1} = 0\}$, so $\Phi(S \cap U) = T \cap V$. Define the transformed pseudopolynomial $g(y, y_r) = \sum_j g_j(y) y_r^{d-j}$, $g_j = f_j \circ \Phi^{-1}$, with analytic coefficients on V .

[changed] *Transport of the witness and of its certificate.* Set $\tilde{P} = P \circ \Phi^{-1}$, analytic on V and not identically zero. Substituting $x = \Phi^{-1}(y)$ in the Bézout identity gives

$$\tilde{P}(y) = \tilde{A}(y, y_r) g(y, y_r) + \tilde{B}(y, y_r) \frac{\partial g}{\partial y_r}(y, y_r), \quad (2)$$

where $\tilde{A}, \tilde{B}$ are polynomials in y_r with coefficients analytic in y (the substitution touches only the base variables, so it commutes with $\partial/\partial x_r$). Thus $\tilde{P} \in \langle g, \partial g/\partial y_r \rangle$ over the ring of analytic functions—with *explicit analytic cofactors*, which the sequel must preserve. By Theorem 2.3.1 (order is preserved by analytic isomorphisms), $\tilde{P}$ is order-invariant in $T \cap V$, with the same constant order $\mu := \text{ord}_0 \tilde{P} < \infty$ as P on S near p . (In the thesis this step defines $E = \text{disc}(g) = D \circ \Phi^{-1}$ instead; no cofactor transport is needed there.)

6.4 Complexification

[unchanged] in method (applied to $\tilde{P}, \tilde{A}, \tilde{B}$ in place of E). Expanding all the analytic coefficient functions (g_j , $\tilde{P}$, and the y -coefficients of $\tilde{A}, \tilde{B}$) in convergent power series at 0 and substituting complex variables (Theorem 2.1.2) extends everything to a polydisc $\Delta_1 \subseteq \mathbb{C}^{r-1}$: a pseudopolynomial $g(z, z_r)$, a holomorphic $\tilde{P}(z)$, and identity (2) continues to hold on Δ_1 by the identity theorem (both sides are holomorphic and agree on the real points, coefficientwise in y_r).

Let $T^* = \{z : z_{s+1} = \cdots = z_{r-1} = 0\}$. The thesis argument on pp. 25–26 applies verbatim to $\tilde{P}$: every partial derivative of $\tilde{P}$ of order $< \mu$ vanishes on $T \cap V$ by order-invariance, its complexification vanishes on $T^* \cap \Delta_1$ (a power series with real coefficients vanishing for real arguments vanishes identically), and some order- μ derivative is nonzero at 0, hence on a polydisc. After refining Δ_1 :

$$\tilde{P}(z) \text{ is order-invariant in } T^* \cap \Delta_1, \text{ with constant (ambient) order } \mu. \quad (3)$$

Moreover the order of the complexification at a real point equals the real order (the complex Taylor coefficients at a real point are determined by the real ones), which is what makes (3) usable and what will later transport the complex conclusions back to $\mathbb{R}$.

6.5 Weierstrass preparation and descent of the membership

Preparation. **[unchanged]** Since $z_r = 0$ is a zero of $g(0, z_r)$ of multiplicity exactly m (and $g(0, z_r) \neq 0$ for $0 < |z_r| \leq \epsilon$), the Weierstrass preparation theorem (Theorem 2.1.10) gives, after refining to $\Delta_2 \subseteq \Delta_1$, a non-vanishing holomorphic unit $u(z, z_r)$ on $\Delta' = \Delta_2 \times \Delta(0; \epsilon)$ and a Weierstrass polynomial

$$h(z, z_r) = z_r^m + a_1(z) z_r^{m-1} + \cdots + a_m(z), \quad a_i(0) = 0,$$

with $g = u h$ on Δ' (thesis eq. (3.3.7)), all roots of $h(z, \cdot)$ lying in $\Delta(0; \epsilon)$.

[changed] *Descent of the membership to $\langle h, h' \rangle$ over germs.* Let $\mathcal{O}$ denote the ring of germs at $0 \in \mathbb{C}^{r-1}$ of holomorphic functions; it is an integral domain (identity theorem), and $h \in \mathcal{O}[z_r]$ is monic of degree m . We claim

$$\tilde{P} \in \langle h, h' \rangle \subseteq \mathcal{O}[z_r] \quad (h' = \partial h / \partial z_r). \quad (4)$$

Indeed, differentiating $g = uh$ in z_r gives $g' = u'h + uh'$; substituting into (2),

$$\tilde{P} = (\tilde{A}u + \tilde{B}u')h + (\tilde{B}u)h',$$

an identity of holomorphic functions of (z, z_r) near 0 whose cofactors $A_1 = \tilde{A}u + \tilde{B}u'$ and $B_1 = \tilde{B}u$ are holomorphic but not polynomial in z_r . To restore polynomiality, divide B_1 by the monic h (Weierstrass division): $B_1 = qh + b$ with $b \in \mathcal{O}[z_r]$ of degree $< m$ and q holomorphic. Then $\tilde{P} - b h' = (A_1 + q h')h$, and the left side is a polynomial in z_r of degree $\leq 2m - 2$ with germ coefficients; dividing by the monic h in $\mathcal{O}[z_r]$ and using that $\mathcal{O}$ is a domain, the holomorphic quotient $A_1 + qh'$ coincides with the polynomial quotient $a \in \mathcal{O}[z_r]$. Hence $\tilde{P} = ah + bh'$ with $a, b \in \mathcal{O}[z_r]$, proving (4)—with the crucial byproduct that the cofactor b has *analytic* coefficients.

6.6 The discriminant of h has constant order along T^*

[changed] (this is the key new step, replacing thesis eq. (3.3.8)). Apply Corollary 4.2 over $R = \mathcal{O}$ to the membership (4):

$$\tilde{P}(z)^m = \text{res}(h, h')(z) \cdot Q(z) = \pm \text{disc}(h)(z) \cdot Q(z), \quad Q = N_{\mathcal{O}[z_r]/(h)}(\tilde{b}) \in \mathcal{O}, \quad (5)$$

valid on a connected open neighbourhood $\Delta_3 \subseteq \Delta_2$ of 0 on which all the germs involved have analytic representatives. The cofactor Q is analytic because the norm is a polynomial expression in the coefficients of b and of h (§4.1).

Neither factor on the right of (5) vanishes identically on Δ_3 : since $\tilde{P} \not\equiv 0$ (it has finite order μ at 0) we have $\tilde{P}^m \not\equiv 0$, and $\mathcal{O}$ is a domain, so $\text{disc}(h) \not\equiv 0$ and $Q \not\equiv 0$. (Note that squarefreeness of f is never invoked: the non-vanishing of $\text{disc}(h)$ is a consequence of the existence of the nonzero witness, cf. Remark 3.2.)

By (3) and additivity of orders, $\text{ord}_z \tilde{P}^m = m\mu$ is constant for $z \in T^* \cap \Delta_3$. Now apply Lemma 4.3 on $U = \Delta_3$ with $S = T^* \cap \Delta_3$ (preconnected), $f = \text{disc}(h)$, $g = Q$:

$$\text{disc}(h) \text{ is order-invariant in } T^* \cap \Delta_3, \text{ with finite constant order.} \quad (6)$$

By Remark 4.4 the lemma applies even though $\text{disc}(h)|_{T^*} \equiv 0$ whenever $m \geq 2$ —the generic situation. Display (6) is precisely the hypothesis of Zariski's theorem.

(For comparison, the thesis obtains (6) from $E = Q \cdot \text{disc}(h)$ with $Q = \text{disc}(u) \text{res}(u, h)^2$ via Theorem 2.3.3, plus Lemma 3.2.2; here the unit u never needs to be analyzed as a pseudopolynomial, and the witness replaces the discriminant E throughout.)

6.7 Application of Zariski's theorem

[unchanged] By (6), Theorem 4.1.1 of the thesis (stated as Theorem 7.1 below) applies to the Weierstrass polynomial h : after refining to $\Delta_4 \subseteq \Delta_3$ there is a holomorphic $\psi : \Delta_4^{(s)} \rightarrow \Delta(0; \epsilon)$ such that for $z = (z_1, \dots, z_s, 0, \dots, 0) \in T^* \cap \Delta_4$ and $\alpha \in \Delta(0; \epsilon)$,

$$h(z, \alpha) = 0 \iff \alpha = \psi(z_1, \dots, z_s) \quad (\text{thesis eq. (3.3.9)}),$$

the unique root having multiplicity exactly m , and h is order-invariant in the graph $G^* = \{(z, \psi(z)) : z \in T^* \cap \Delta_4\}$. This delivers both the *nonsplitting* of the root cluster over the section and the order-invariance of h (hence of $g = uh$, as $u \neq 0$) along it.

6.8 Return to the real domain and reassembly

[unchanged] The argument of thesis pp. 27–28 applies verbatim:

1. For real $y \in T \cap B$ the polynomial $g(y, y_r)$ has real coefficients, so the unique root $\psi(y)$ of $h(y, \cdot)$ in $\Delta(0; \epsilon)$ is fixed by conjugation, i.e. real. Consequently the power series of ψ has real coefficients and the restriction $\eta = \psi|_{B(s)}$ is a real-valued real-analytic function (Schwarz reflection, via Theorem 2.1.2).
2. For $y \in T \cap B$, $\alpha \in (-\epsilon, \epsilon)$: $g(y, \alpha) = 0 \iff h(y, \alpha) = 0 \iff \alpha = \eta(y)$, since $u \neq 0$.
3. The order of g at a real point of the graph of η equals the order of its complexification at that point (matching Taylor coefficients, as in §6.4); hence the order-invariance of h in G^* gives order-invariance of g in the graph of η .
4. Detransform: $N = \Phi^{-1}(B)$ and $\theta = \eta \circ \phi$, where ϕ is the chart of S induced by Φ . By Theorem 2.3.1 the order is preserved, establishing Assertion 2 for the root α_i .

Undoing the translation of §6.1 and intersecting the finitely many neighbourhoods N_i yields Assertion 1 at p ; globalization over the connected S was already described in §6.1. Finally, for the conclusion as stated (order-invariance along *arbitrary* continuous root functions θ): on each connected local piece, the values of θ lie among the finitely many separated analytic branches, so the branch index is locally constant, hence constant; thus θ coincides with a single delineation branch and inherits its order-invariance. $\square$

7 The Zariski theorem (thesis Chapter 4): unchanged

The generalization makes *no* change to Chapter 4 of the thesis: the hypothesis it consumes, constancy of the order of $\text{disc}(h)$ along the section, is produced by §6.6, and its conclusion is consumed in §6.7 exactly as in the original proof. We restate it for completeness.

Theorem 7.1 (Zariski; thesis Theorem 4.1.1). *Let $h(z, z_n)$ be a Weierstrass polynomial of degree $m \geq 1$ on a polydisc $\Delta_1 \subseteq \mathbb{C}^{n-1}$, all of whose roots lie in $\Delta(0; \epsilon)$, and whose discriminant $F(z)$ does not vanish identically. Let $1 \leq s \leq n-2$ and $T^* = \{z : z_{s+1} = \dots = z_{n-1} = 0\}$, and assume F is order-invariant in $T^* \cap \Delta_1$. Then on a smaller polydisc there is a holomorphic $\psi(z_1, \dots, z_s)$ with values in $\Delta(0; \epsilon)$ such that, over points of T^* , the roots of h are exactly $z_n = \psi$ (“nonsplitting”: one root of multiplicity m), and h is order-invariant in the graph G^* of ψ over T^* .*

Since the formalization had to be axiom-free, the entire Chapter-4 proof was formalized as well, following the thesis: the splitting of Theorem 4.2.1 into the nonsplitting statement (Theorem 4.2.2) and the order-invariance statement (Theorem 4.2.3); factorization of h into irreducible Weierstrass polynomials over the germ ring; the root cover of the separable locus and path lifting (Lemma 4.2.4); transitive monodromy for irreducible factors (Lemma 4.2.5); the homotopy-deformation contradiction proving nonsplitting per factor and the resultant/zero-set argument making the factors’ roots coincide (proof of Theorem 4.2.2); the Newton–Puiseux parametrization $z_{n-1} = u^m$, $z_n = \phi(z, u)$ (Lemma 4.2.6) and the order computation $\text{ord}_{(z,0,0)} h = \min(m, m_1)$ (Lemmas 4.2.7–4.2.8, proof of Theorem 4.2.3); and, for codimension ≥ 2 , the quadratic blow-up $Q(z, z_{s+1}, \dots, z_{n-1}) = (z, z_{s+1}z_{n-1}, \dots, z_{n-2}z_{n-1}, z_{n-1})$ reducing to codimension one, including the normal form $F \circ Q = z_{n-1}^r N$ with $N(0) \neq 0$ (stages 1–7 of the thesis proof of 4.1.1) and the term-tracking transport of the order of h through the blow-up (stage 8). A few proofs were reorganized in the formalization without changing the statements; the noteworthy points are:

- *Lemma 4.2.5.* The thesis cites Bochner–Martin for the transitive monodromy of the root cover of an irreducible Weierstrass polynomial. The formalization proves it: a clopen splitting of the root cover over the punctured (separable) base would glue, via Cauchy-coefficient

continuity and a removable-singularity argument, into a nontrivial monic factorization of h over the germ ring, contradicting irreducibility; path-connectedness of the cover then yields transitive monodromy by lifting.

- *Genericity of the blow-up direction* (stage 1 of the thesis proof): the linear change making $\partial^r F / \partial z_{n-1}^r(0) \neq 0$ is obtained from a polarization argument—the degree- r leading form of F is a nonzero symmetric form, so its diagonal restriction cannot vanish on any dense set of transverse directions. The same mechanism drives the stage-8 term-tracking (the nonzero coefficient q of the thesis’s displayed term is realized at some nearby ratio point w'').
- *Lemmas 4.2.7–4.2.8*. Where the thesis works in iterated fractional-power-series fields, the formalization stays with the convergent parametrization of Lemma 4.2.6 and uses only *continuity* of Cauchy-integral coefficients in the section variables, which suffices for the order computation $\text{ord}_{(z,0,0)} h = \min(m, m_1)$.
- The auxiliary identities of Theorem 2.3.3 (discriminant of a product) *are* used here—inside Chapter 4, to descend discriminant order-invariance from h to its irreducible factors and to force distinct factors to share their section root—exactly as in the thesis proofs of Theorems 4.2.2 and 4.2.3.

We emphasize the division of labour this creates for the generalized theorem: all uses of the *specific* polynomials *disc*, *res* in McCallum’s Chapter 3 argument are eliminated in favour of the elimination-ideal witness, while their uses *internal* to the Zariski theorem are untouched—they concern the Weierstrass polynomial h and its factors, objects private to the proof, not the projection set handed to the CAD algorithm.

8 Summary of changes, and the Lean formalization

8.1 What changed relative to the thesis

Thesis item	Status	Notes
Ch. 2 (preliminaries)	unchanged	used as is
Def. of $P(A)$	changed	$P'(A)$ with chosen elimination-ideal elements (Def. 3.1)
Thm. 3.2.1 statement	changed	witness $P \in \langle f, f' \rangle \cap \mathbb{R}[x]$ replaces D ; squarefree/ $D \neq 0$ subsumed by $P \neq 0$
Thm. 3.2.3 statement	changed	$P'(A)$ replaces $P(A)$
Proof of 3.2.3 from 3.2.1	changed	radical lemma (Lemma 4.5) replaces $D = \prod \text{disc} \cdot \prod \text{res}^2$ (Thm. 2.3.3)
3.2.1: reduction to Assertions 1, 2; root clusters; translation	unchanged	§6.1
3.2.1: case $s = r - 1$	simplified	$P(a) \neq 0 \Rightarrow$ separability directly (§6.2)
3.2.1: coordinate change	unchanged	§6.3
3.2.1: transform of the invariant	changed	$\tilde{P} = P \circ \Phi^{-1}$ with <i>Bézout cofactors</i> replaces $E = \text{disc } g$
3.2.1: complexification	unchanged in method	applied to $\tilde{P}$ (and cofactors) instead of E
3.2.1: Weierstrass preparation	unchanged	eq. (3.3.7)
3.2.1: membership descent to $\langle h, h' \rangle$	new	Weierstrass division of the cofactors (§6.5)
3.2.1: factorization step (3.3.8)	changed	norm identity $\tilde{P}^m = \pm \text{disc}(h) Q$ (Cor. 4.2) + Lemma 4.3 replace Thm. 2.3.3
3.2.1: Zariski application (3.3.9)	unchanged	§6.7
3.2.1: return to real; reassembly	unchanged	§6.8
Ch. 4 (Zariski theorem)	unchanged	statement and proof as in thesis; formalized in full (§7)
[new] norm-resultant identity	new	Lemma 4.1, Cor. 4.2 (classical algebra)
[new] order-additivity lemma	new	Lemma 4.3 = ambient “only if” half of Lemma 3.2.2, full proof
[new] radical lemma	new	Lemma 4.5 (pure ring theory)

8.2 The formalization

The entire development is machine-checked in Lean 4 over Mathlib. The main theorem is `mccallum_3_2_3_generalized` (`Mccalum/Generalized/Projection.lean`); `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`. In particular, the classical inputs that the informal literature cites—the Weierstrass preparation and division theorems for convergent power series (absent from Mathlib), the several-variable holomorphic $\Rightarrow$ analytic bridge needed for them, Zariski’s theorem with its monodromy and blow-up arguments—are all proved inside the development. A correspondence table for the main steps:

This document	Lean identifier (file)
Theorem 3.5 (3.2.3')	mccallum_3_2_3_generalized (Generalized/Projection.lean)
Theorem 3.3 (3.2.1')	lifting_theorem_generalized (ibid.)
Lemma 4.5	nullstellensatz_elim (ibid.)
Lemma 4.1	norm_eq_resultant_mononic (Generalized/Lifting.lean)
Corollary 4.2	norm_identity_elim (ibid.)
Lemma 4.3	order_factor_const_of_mul_analytic (OrderMulAnalytic.lean)
§6.2 (open case)	lifting_generalized_open_case, separable_of_elim_nonvanishing (Generalized/Lifting.lean)
§6.3 (chart + transport)	lifting_generalized_codim_local (Generalized/Delineable.lean)
§6.4	analyticAt_complexify, order_real_eq_order_complex (Generalized/Lifting.lean)
§6.5 (preparation; descent)	weierstrass_preparation_analytic (Generalized/WeierstrassPrep.lean); descent_membership (Generalized/MembershipDescent.lean)
Eq. (5) over germs	descent_norm_identity (Generalized/DescentNormIdentity.lean)
§6.6 (disc order constant)	weierstrassDisc_order_const_along_section (Generalized/DiscOrder.lean)
Theorem 7.1 as consumed	cluster_root_structure (Generalized/ClusterRootStructure.lean) = zariski_single_branch (Conclusion 1, Generalized/ZariskiNonsplitting.lean) + Puisseux.order_invariant_in_graph (Conclusion 2, Puisseux/Conclusion2General.lean)
Thesis Thm. 4.2.2 (nonsplitting, codim 1)	irreducible_section_single_root_e1 (Generalized/ZariskiE1.lean); monodromy kernel in Generalized/Monodromy*.lean, Puisseux/RootCover.lean
Thesis Lemma 4.2.5	rootCover_exchange_gen (Generalized/MonodromyGen.lean)
Thesis Thm. 4.2.3 / Lemmas 4.2.6–4.2.8	Puisseux/Parametrization.lean, Puisseux/OrderInvariance.lean, Puisseux/Conclusion2Value*.lean (order_eval_eq_min)
Thesis 4.1.1 Case II (blow-up; stage 8)	ZariskiE2.irreducible_section_single_root_blowup_proof (Generalized/ZariskiE2Main.lean); Puisseux.order_eval_value_e2_blowup_proof (Puisseux/Conclusion2E2.lean)
§6.8, reassembly, globalization	Generalized/ClusterFromReal.lean, MultiCluster.lean; locally_delineable_to_global', order_invariant_of_locally_invariant (Generalized/Delineable.lean)

Two findings of the formalization are worth recording. First, the natural strengthening “ $\langle h, g \rangle \cap R = (\text{res}(h, g))$ for monic h ” that one might be tempted to use in §6.6 is false (counterexample (1)); the formal proof effort exposed this early, and Lemma 4.1 is the correct replacement. Second, the deduction of the projection theorem needs no Nullstellensatz and no

algebraically closed field: the radical-as-intersection-of-primes argument of Lemma 4.5 is both simpler to formalize and sharper than the geometric argument it replaces.

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